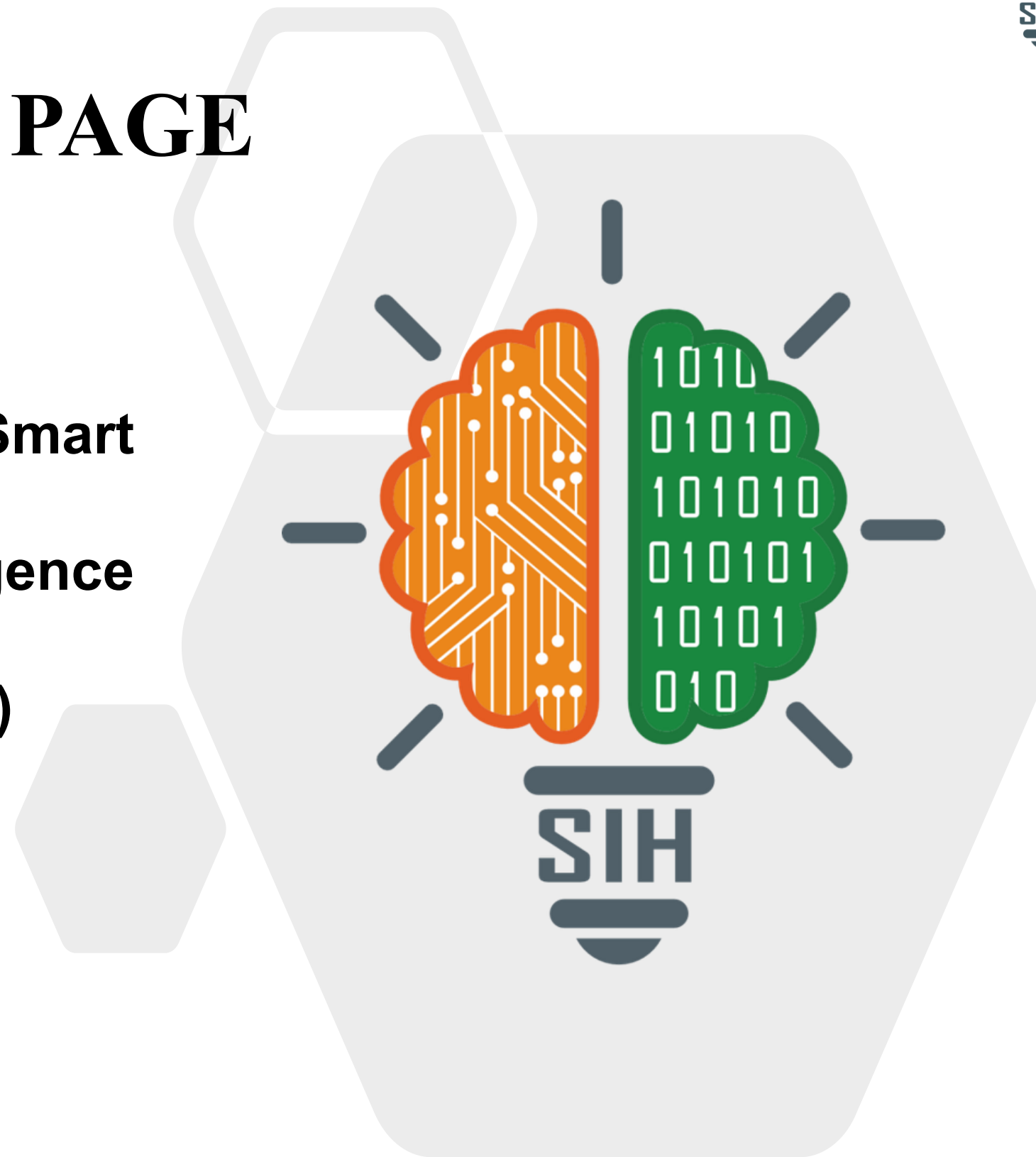
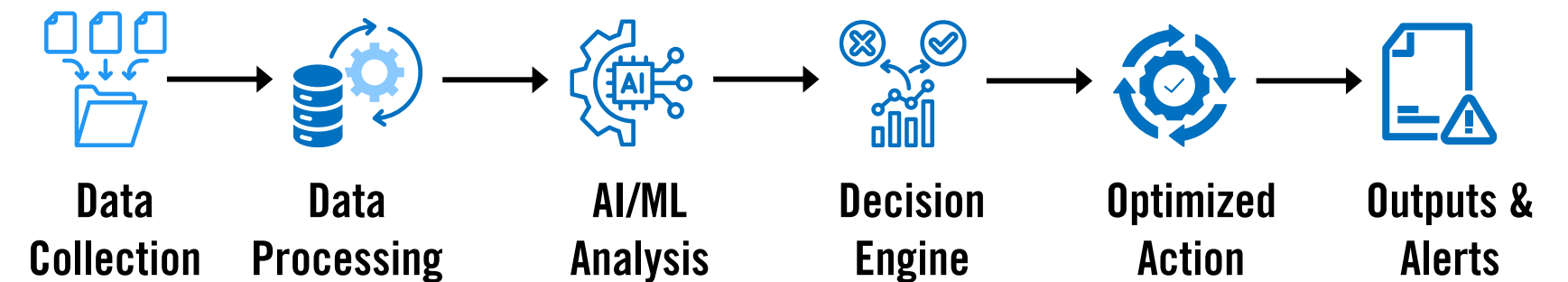


TITLE PAGE

- Problem Statement ID – SIH26002
- Problem Statement Title – AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
- Theme - Transportation and Logistics
- PS Category - Software
- Team ID - 160406
- Team Name – LogiHawks



System Workflow



DATA SOURCE



GPS & IOT
Devices



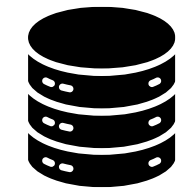
Weather
APIs



Traffic
Data



Maps & GIS
Data



Govt. & Public
Database

WHO IS AFFECTED ?



Businesses



Farmers



Tourists



Transporters

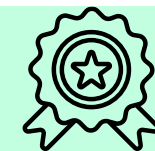


Govt. &
Services

SOLUTION

AI-Powered platform for smart logistics & accessibility intelligence.

- ✓ Real -time multi-modal logistics tracking.
- ✓ AI-driven route optimization.
- ✓ Weather & disruption prediction.
- ✓ Accessibility mapping for remote areas.
- ✓ Dashboard for buisness & authorities.
- ✓ Promotes efficient, reliable & inclusive connectivity in NER.



Key Feature:



AI-based route optimization.



Real-time vehicle & cargo tracking.



Weather & landslide alerts.



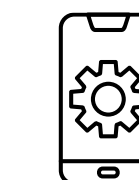
Accessibility & connectivity mapping.



Demand forecasting & analytics.



Multi-modal integration (Road,Rail,Air,Water) .



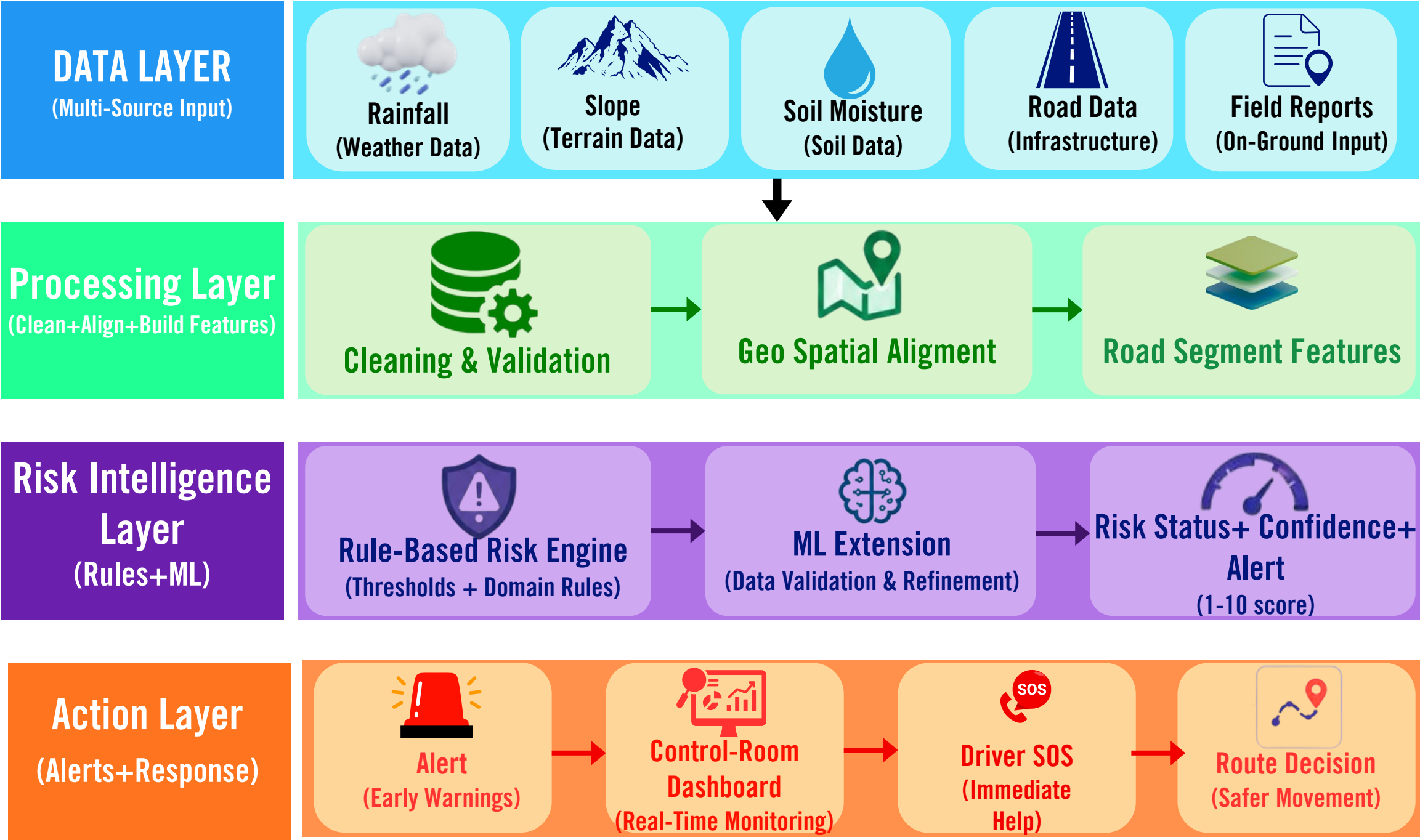
User-friendly dashboards & mobile app.

Technical Stack

- Frontend: React.js, Next.js, Leaflet
- Backend: Python, Fast-API
- Database: SQL-lite, Post-GIS
- Offline SOS: Local Storage + Peer Relay
- Routing Engine: Future Extension
- Optional Hardware Extension: LoRa vehicle, Roadside node



Methodology & Implementation Flow



FEASIBILITY

- ✓ Technically feasible with existing technologies.
- ✓ Cost-effective & scalable solution.
- ✓ High availability of data sources (API's, GIS)
- ✓ Strong government support for digital transformation.



SECURITY & PRIVACY

- ✓ Secure Authentication
- ✓ Data Encryption
- ✓ Role-Based Access Control
- ✓ Privacy Compliance
- ✓ Cybersecurity Protocols



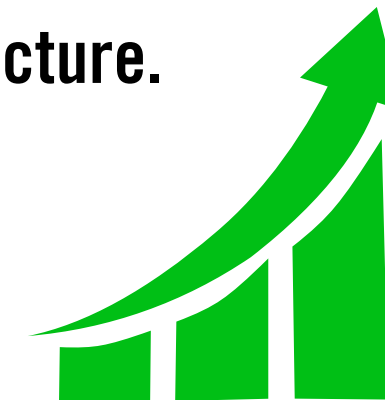
FUTURE SCOPES

- Expansion to all NER states & cross-border trade.
- Smart shipment corridors & EV logistics.



SCALABILITY

- ✓ Cloud-base scalable infrastructure.
- ✓ Can onboard more vehicles, users & data.
- ✓ Expandable to other regions & countries.
- ✓ Future-ready architecture.

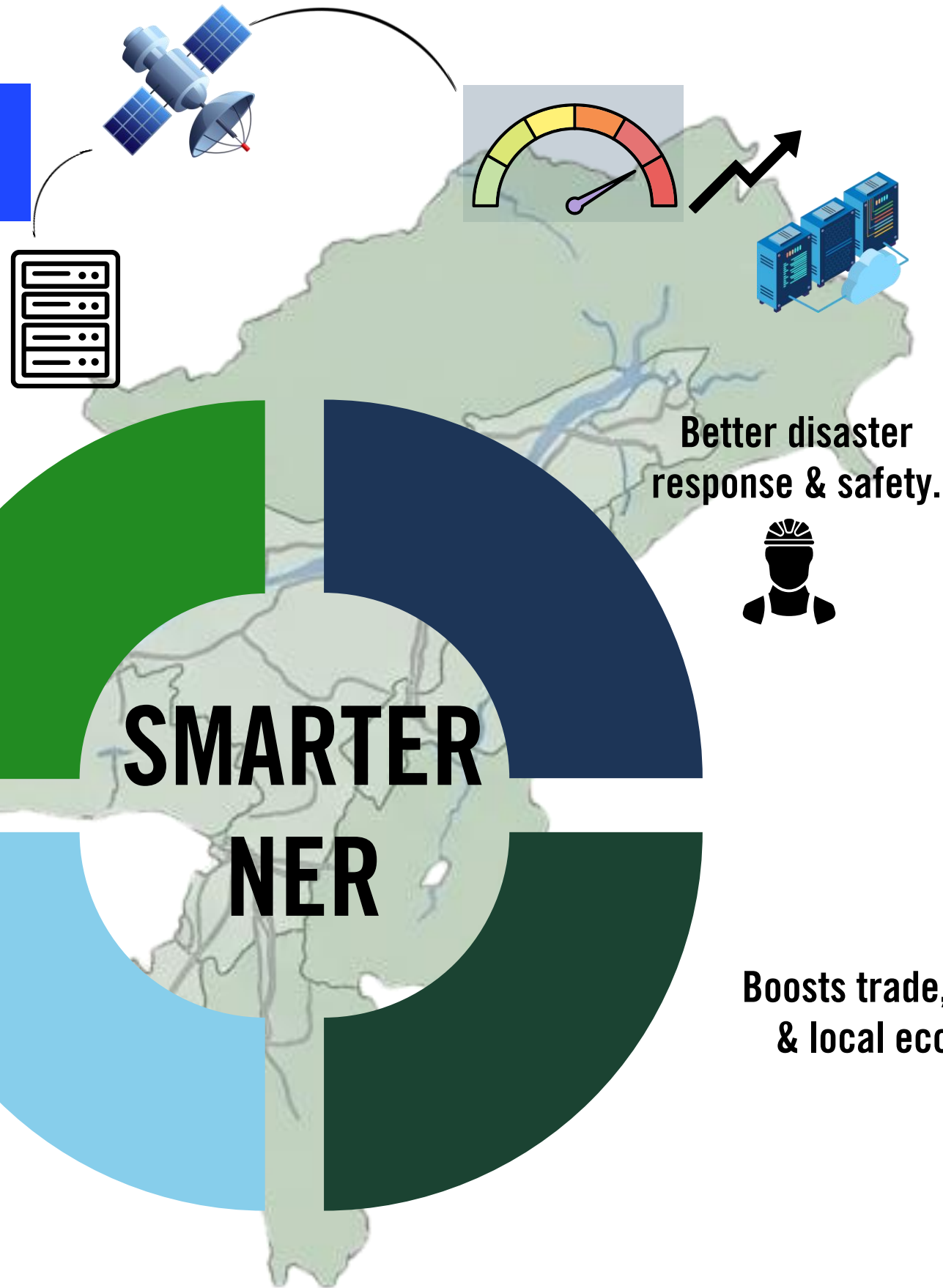


PROTOTYPE (MVP)

- Live Route Tracking
- AI-driven Route Suggestions.
- Weather & Alert Notifications.
- Dashboard Analytics.



IMPACTS



BENEFICIARIES

-  Local Businesses & Industries.
-  Farmers & Producers.
-  Transport & Logistics Companies.
-  Tourists & Travelers.
-  Government & Policy Makers

EXPECTED RESULTS

- 20%-30% reduction in logistics cost..
- 30%-40% improvement in delivery time.
- Real-time visibility & transparency.
- Better route safety & reliability.
- Increased accessibility index for NER.

RESEARCH BASIS

Policy & Problem: SIH26002 (MoDNER), Bharatmala (MoRTH/PIB)

[Home](#) | [NDMA](#)

Hazard & Weather: IMD, NDMA, GSI, NESAC

<https://gsi.gov.in/>

Geospatial & Road: ISRO Bhuvan, OSM, MoRTH Black Spot Portal

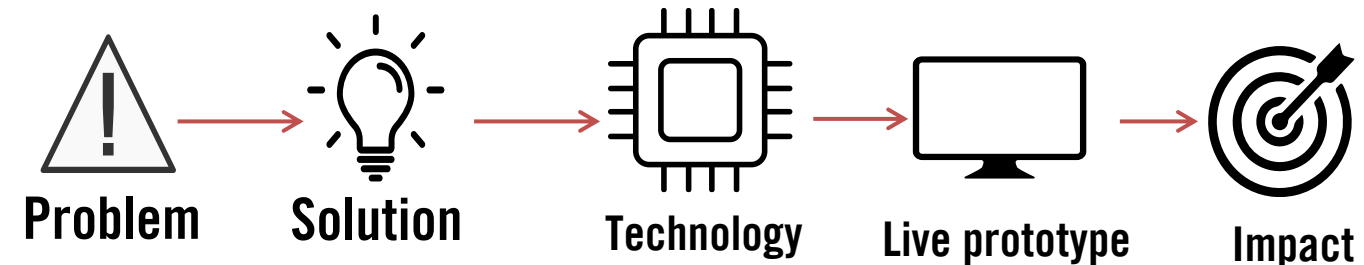
<https://openstreetmap.org>

Communication: Google Nearby, Wi-Fi Aware, DoT 865–868 MHz

REFERENCES

- Govt. Reports & NER Statistics.
- Research Papers on AI in Logistics.
- Open Datasets (GIS, Weather, Traffic).
- APIs (Google Maps, Open Weather)
- Existing Solutions & Case Studies.

Final Demo

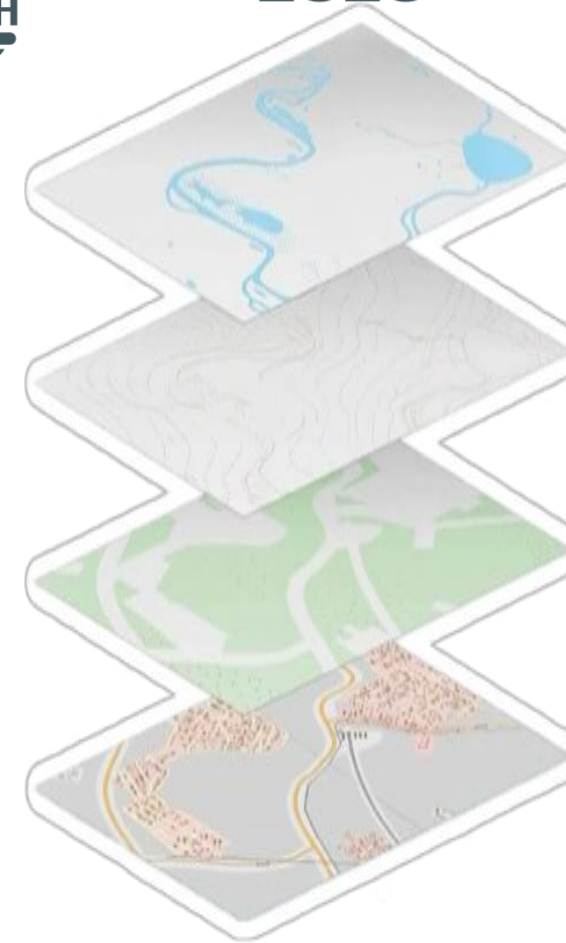


Water bodies map

Elevation map

Vegetation map

Infrastructure map



https://youtu.be/9unA_tUA50I



[GITHUB REPO OF MARGSETU](#)